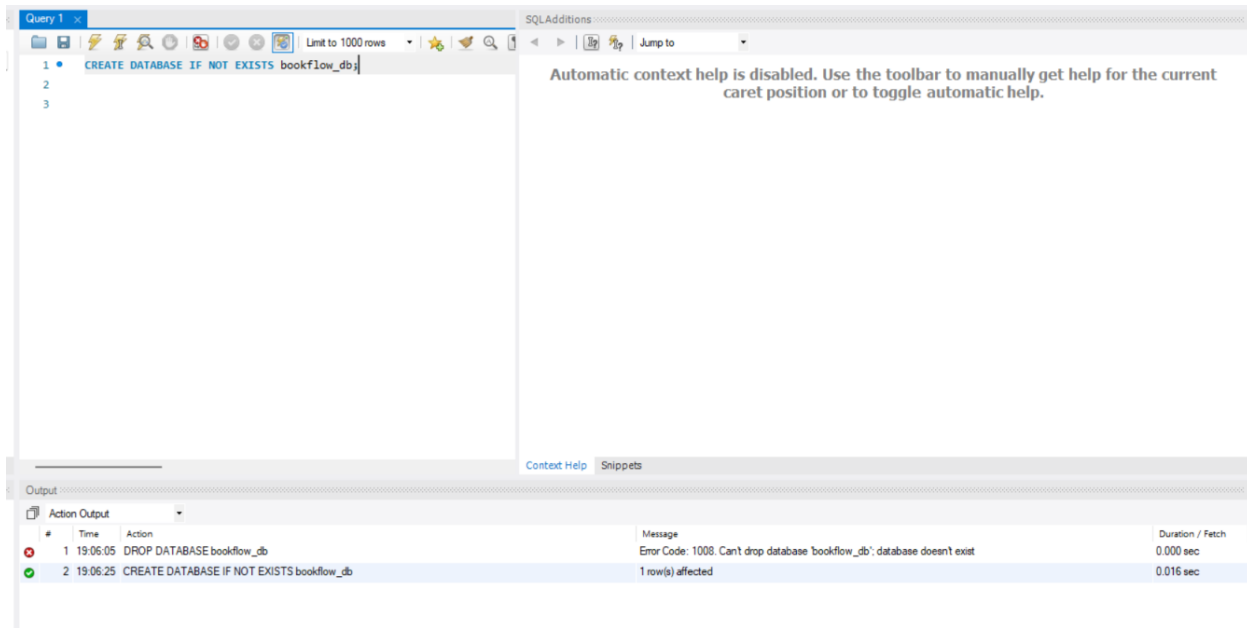


Database System Engineering and Distributed Backend Development

Week-1



The screenshot shows a SQL IDE interface with a query editor on the left and a results/output pane on the right. The query editor contains the following SQL statement:

```
1 CREATE DATABASE IF NOT EXISTS bookflow_db;
2
3
```

The results/output pane displays a message indicating that automatic context help is disabled. Below this, the 'Output' section shows a table with the following data:

#	Time	Action	Message	Duration / Fetch
1	19.06.05	DROP DATABASE bookflow_db	Error Code: 1008. Can't drop database 'bookflow_db'; database doesn't exist	0.000 sec
2	19.06.25	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.016 sec

4

5 • CREATE TABLE books (
6 book_id INT AUTO_INCREMENT PRIMARY KEY,
7 title VARCHAR(255) NOT NULL,
8 isbn VARCHAR(13) NOT NULL UNIQUE,
9 published_year INT,
10 CONSTRAINT chk_published_year CHECK (published_year < 2027)
11);
12

13 • CREATE TABLE members (
14 member_id INT AUTO_INCREMENT PRIMARY KEY,
15 full_name VARCHAR(100) NOT NULL,
16 email VARCHAR(150) NOT NULL UNIQUE
17);
18
19
20

Output

Action Output

#	Time	Action	Message
✓ 93	19:20:51	USE bookflow_db	0 row(s) affected
✓ 94	19:20:51	DROP DATABASE IF EXISTS bookflow_db	1 row(s) affected
✓ 95	19:21:17	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 96	19:21:18	USE bookflow_db	0 row(s) affected
✓ 97	19:21:18	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT N...	0 row(s) affected
✓ 98	19:21:18	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(1...	0 row(s) affected

1 • CREATE DATABASE IF NOT EXISTS bookflow_db;
2 • USE bookflow_db;
3 • CREATE TABLE books (
4 book_id INT AUTO_INCREMENT PRIMARY KEY,
5 title VARCHAR(255) NOT NULL,
6 isbn VARCHAR(13) NOT NULL UNIQUE,
7 published_year INT,
8 CONSTRAINT chk_published_year CHECK (published_year < 2027)
9);
10
11

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	19:06:25	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.016 sec
⚠ 3	19:07:47	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
✓ 4	19:07:47	USE bookflow_db	0 row(s) affected	0.000 sec
⚠ 5	19:08:01	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
✓ 6	19:08:01	USE bookflow_db	0 row(s) affected	0.000 sec
✓ 7	19:08:01	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT N...	0 row(s) affected	0.062 sec

Query 1

```

13 • CREATE TABLE members (
14     member_id INT AUTO_INCREMENT PRIMARY KEY,
15     full_name VARCHAR(100) NOT NULL,
16     email VARCHAR(150) NOT NULL UNIQUE
17 );
18 • DESCRIBE books;
19 • DESCRIBE members;

```

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI		auto_increment
full_name	varchar(100)	NO			
email	varchar(150)	NO	UNI		

Result 1

```

8     isbn VARCHAR(13) NOT NULL UNIQUE,
9     published_year INT,
10    CONSTRAINT chk_published_year CHECK (published_year < 2027)
11 );
12
13 • CREATE TABLE members (
14     member_id INT AUTO_INCREMENT PRIMARY KEY,
15     full_name VARCHAR(100) NOT NULL,
16     email VARCHAR(150) NOT NULL UNIQUE
17 );
18 • DESCRIBE books;
19 • DESCRIBE members;
20 • INSERT INTO books (title, isbn, published_year) VALUES
21     ('The Alchemist', '9780061122415', 1988),
22     ('Clean Code', '9780132350884', 2008),
23     ('Atomic Habits', '9780735211292', 2018);
24
25
26

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
97	19:21:18	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT N...	0 row(s) affected	0.062 sec
98	19:21:18	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(...	0 row(s) affected	0.063 sec
99	19:23:25	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
100	19:23:46	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
101	19:24:57	DESCRIBE members	3 row(s) returned	0.000 sec / 0.000 sec
102	19:25:52	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

```

27      ('Ravi Verma', 'ravi.verma@example.com');
28 • SELECT * FROM books;

```

Result Grid				
Filter Rows:				
Edit: Export/Import:				
	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
•	NULL	NULL	NULL	NULL

```

29 • SELECT * FROM members;

```

Result Grid

Filter Rows:

Edit: Export/Import:

Wrap Cell Content: ☐

Result Grid

Form Editor

Field Types

members 3 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
102	19:25:52	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
103	19:26:33	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Shama', '...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.046 sec
104	19:26:39	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
105	19:26:42	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Shama', '...	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'	0.016 sec
106	19:27:23	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
107	19:27:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

```

30 • INSERT INTO books (title, isbn, published_year)
31   VALUES ('Fake Copy', '9780061122415', 2000);
32 • INSERT INTO books (title, isbn, published_year)
33   VALUES (NULL, '9999999999999', 2010);
34
35
36

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105	19:26:42	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Shama', '...	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'	0.016 sec
106	19:27:23	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
107	19:27:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
108	19:28:49	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.015 sec
109	19:29:37	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
110	19:29:40	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null	0.000 sec

```
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Time Traveler', '888888888888', 2030);
36 • INSERT INTO members (full_name, email)
37 VALUES ('Anil Clone', 'anil.kumar@example.com');
```

38

[Context Help](#) [Snippets](#)

Output



Action Output

#	Time	Action	Message	Duration / Fetch
✓ 107	19:27:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✗ 108	19:28:49	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.015 sec
✗ 109	19:29:37	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
✗ 110	19:29:40	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null	0.000 sec
✗ 111	19:30:22	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
✗ 112	19:30:26	INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'	0.047 sec